

INFLUENCE OF ENSO IN INDIAN AND SRI LANKAN MONSOON RAINFALL

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Term Project Report - III
Trimester 9

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ABSTRACT

This study looks at how the El Niño–Southern Oscillation (ENSO) affected rainfall across Sri Lanka and India from 1996 to 2025. It then applies the findings to the ongoing El Niño event in 2026. Rather than being restricted to a single presumed monsoon season, four zones such as, India Wet (Western Ghats), India Dry (Deccan), Sri Lanka Wet (South West), and Sri Lanka Dry (North East), were analyzed independently using the Oceanic Niño Index (ONI) and gridded CHIRPS precipitation. Correlation was examined at monthly resolution throughout the entire annual cycle. This monthly-resolved analysis shows that both countries have real, statistically significant ENSO sensitivity, but it is concentrated in discrete, physically separate times. India’s signal is strongest in August–October (Dry zone August: $r = -0.562$, $p = 0.001$); Sri Lanka shows a strong positive (wetter) signal in October ($r = +0.594$, $p = 0.0005$, the strongest single correlation in this study) followed by a significant drying signal spanning February–April ($r = -0.37$ to -0.51 across zones and months). The monthly decomposition reveals that the previous JJAS-only analysis’s conclusion that Sri Lanka had no ENSO sensitivity was an artifact of testing the incorrect season rather than a lack of a true link. Applying these relationships to the current event – an El Niño Advisory as of August 2026, with Niño 3.4 forecast to reach $+3.4^{\circ}\text{C}$ by October and a 97% chance of persistence through early spring 2027 – projects a substantially below-normal remainder of India’s 2026 monsoon (most acute in August, -3.5 to -4.2 mm/day), an anomalously wet October 2026 for Sri Lanka ($+3$ to $+8$ mm/day, conservatively capped at the historical maximum observed ONI to avoid extrapolation beyond the historical record), and a multi-month drought risk for Sri Lanka spanning February–April 2027 (-1.5 to -4.1 mm/day). These projections are broadly consistent with, and provide historical statistical grounding for, public advisories issued by both governments regarding this event.

Index Terms— El Niño, ENSO, Oceanic Niño Index, Indian Monsoon, CHIRPS, Precipitation Variability

1. INTRODUCTION

The El Niño–Southern Oscillation (ENSO) is the dominant mode of interannual climate variability on Earth, arising from coupled ocean–atmosphere interactions in the equatorial Pacific. Its warm phase, El Niño, and cool phase, La Niña, are known to influence weather patterns far beyond the Pacific basin through atmospheric teleconnections. Among these, the relationship between ENSO and the South Asian summer monsoon is one of the most extensively studied, given

the monsoon’s central importance to agriculture, water resources, and the economies of India and Sri Lanka.

Understanding how strongly, and under what conditions, ENSO modulates monsoon rainfall is important both for seasonal forecasting and for anticipating drought or flood risk in the region. While the broad relationship (El Niño tends to suppress monsoon rainfall) is well established in the climate literature, the strength of this relationship can vary considerably by sub-region, depending on local topography and rainfall regime.

This project quantifies the ENSO–monsoon relationship over India and Sri Lanka for the period 1996–2025, using the Oceanic Niño Index (ONI) as the ENSO indicator and gridded CHIRPS precipitation data as the rainfall record. The region is split into a wet zone (Western Ghats) and a dry, rain-shadow zone (Deccan Plateau) to test whether the strength of the relationship differs between areas with strong orographic rainfall and areas more directly dependent on large-scale monsoon circulation.

Section 2 describes the problem formulation and objectives. Section 3 details the methodology, including data sources and event-detection approach. Experimental results are presented in Section 4, followed by discussion and conclusions in Sections 5 and 6.

2. PROBLEM STATEMENT AND OBJECTIVES

2.1. Problem Statement

El Niño events are known to influence global weather patterns, but the strength and consistency of this influence over the Indian/Sri Lankan monsoon region, and whether it varies between the two countries and between wetter and drier sub-regions within each, requires direct quantitative verification against observational data rather than being assumed from the general literature alone. This project addresses two related questions: *does the Oceanic Niño Index show a statistically significant relationship with monsoon-season rainfall over India and Sri Lanka between 1996 and 2025, and does the strength of this relationship differ (i) between India and Sri Lanka, and (ii) between a wet and dry zone within each country?*

2.2. Objectives

1. To identify and classify El Niño, La Niña, and Neutral periods over 1996–2025 using the Oceanic Niño Index (ONI).
2. To obtain and process gridded precipitation data (CHIRPS) over India and Sri Lanka, and to define four representative sub-regions: a wet and dry zone within each country, kept

separate rather than merged, so that national-level differences can be examined alongside zonal ones.

3. To quantify the correlation between ONI and monsoon-season (JJAS) rainfall anomalies in each of the four zones, including statistical significance testing.
4. To compare mean rainfall anomalies across ENSO phases (El Niño, La Niña, Neutral) using analysis of variance (ANOVA), for each zone.
5. To determine whether India or Sri Lanka shows stronger ENSO sensitivity, and to interpret the physical basis for any observed difference.

3. METHODOLOGY / APPROACH

This analysis was adapted from a group project originally conducted for Madagascar, which examined the relationship between ENSO, precipitation, and vegetation (NDVI) over the Southwest Indian Ocean region. The same overall pipeline – ONI-based event detection, gridded precipitation retrieval, zonal clipping, and phase-stratified correlation analysis – was re-applied to the India/Sri Lanka monsoon region as an independent extension, given the well-documented and physically distinct ENSO–monsoon teleconnection in this region.

3.1. Overall Workflow

The workflow proceeds in four stages: (1) ONI data acquisition and El Niño event classification, (2) definition of the India/Sri Lanka study region and its four sub-zones, (3) retrieval and spatial clipping of CHIRPS precipitation data, and (4) statistical analysis relating ONI to zonal rainfall anomalies.

3.2. Technical Details

ENSO classification. The Oceanic Niño Index (ONI) [1], a 3-month running mean of sea surface temperature anomalies in the Niño 3.4 region, was obtained from NOAA’s Climate Prediction Center for 1996–2025. Following NOAA’s operational definition, a period was classified as an El Niño (La Niña) event when ONI remained at or above $+0.5^\circ\text{C}$ (at or below -0.5°C) for at least five consecutive overlapping 3-month seasons; all other periods were classified as Neutral.

Precipitation data. Daily gridded precipitation at 0.25° resolution was obtained from the CHIRPS v2.0 dataset [2] (Climate Hazards Group, UC Santa Barbara) for 1996–2025 and clipped to the India and Sri Lanka national boundaries (Natural Earth administrative boundaries [3]).

Zone definition. Four sub-regions were defined to represent contrasting rainfall regimes within each country, kept separate rather than merged across countries: an *India wet zone* covering the Western Ghats and Kerala coastal belt, an *India dry zone* covering the Deccan Plateau rain-shadow region, a *Sri Lanka wet zone* covering the southwest of the island, and a *Sri Lanka dry zone* covering the north, east, and southeast of the island. Each zone was clipped to its respective country’s national boundary independently, rather than to a combined India–Sri Lanka boundary, so that country-level comparisons would not be confounded by cross-border averaging.

Statistical analysis. Monthly rainfall anomalies were computed relative to each month’s 1996–2025 climatology for both zones. Pearson correlation coefficients between ONI and rainfall anomaly were computed (i) across all months and (ii) restricted to the monsoon season (June–September, JJAS), since the ENSO–monsoon

teleconnection is expected to be strongest during this period. One-way ANOVA was used to test whether mean JJAS rainfall anomalies differed significantly across the three ENSO phases (El Niño, La Niña, Neutral).

3.3. Mathematical Formulation

ONI and rainfall anomaly. ONI is the 3-month running mean of sea surface temperature anomaly in the Niño 3.4 region relative to a 1991–2020 climatology, classified as El Niño ($\geq +0.5^\circ\text{C}$) or La Niña ($\leq -0.5^\circ\text{C}$) for at least 5 consecutive overlapping seasons, and Neutral otherwise. For each zone, the rainfall anomaly $A_t = P_t - \bar{P}_{c(t)}$, where P_t is zone-averaged monthly precipitation and $\bar{P}_{c(t)}$ is the climatological mean for calendar month $c(t)$.

Pearson correlation. The linear ONI–rainfall relationship is quantified by

$$r = \frac{\sum_i (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_i (x_i - \bar{x})^2} \sqrt{\sum_i (y_i - \bar{y})^2}} \quad (1)$$

with $x_i = \text{ONI}_i$, $y_i = A_i$; significance assessed via a two-sided t -test with $n - 2$ degrees of freedom.

Linear regression. To translate a given ONI value into a projected rainfall anomaly (used for the current-event outlook in Section 5), an ordinary least-squares regression is fit for each zone and month/season of interest:

$$\hat{A} = \beta_1 \cdot \text{ONI} + \beta_0 \quad (2)$$

where the slope and intercept minimize the sum of squared residuals,

$$\beta_1 = \frac{\sum_i (x_i - \bar{x})(y_i - \bar{y})}{\sum_i (x_i - \bar{x})^2}, \quad \beta_0 = \bar{y} - \beta_1 \bar{x} \quad (3)$$

again with $x_i = \text{ONI}_i$ and $y_i = A_i$ restricted to the relevant month(s). Predictions made at an input ONI value outside the range of x_i used to fit the model constitute extrapolation rather than interpolation, and are flagged as such wherever they occur in this report (Section 5).

One-way ANOVA. To test whether mean anomaly differs across phases $k \in \{\text{El Niño, La Niña, Neutral}\}$:

$$F = \frac{\sum_k n_k (\bar{A}_k - \bar{A})^2 / (K - 1)}{\sum_k \sum_{i \in k} (A_i - \bar{A}_k)^2 / (N - K)} \quad (4)$$

where $\bar{A}_k$, n_k are the mean and count in phase k , $\bar{A}$ the overall mean, $K = 3$, and N the total sample size. A large F indicates phase means differ more than expected from within-phase variability alone.

4. EXPERIMENTS AND RESULTS

4.1. Experimental Setup

The analysis used ONI data from NOAA CPC (1996–2025, monthly/seasonal resolution) and CHIRPS v2.0 daily precipitation data (0.25° resolution, aggregated to monthly means) over the same period. All processing was performed in Python using `xarray`, `rioxarray`, `geopandas`, and `scipy.stats`. Statistical significance was assessed at the conventional $p < 0.05$ threshold, with $p < 0.01$ and $p < 0.001$ noted where applicable.

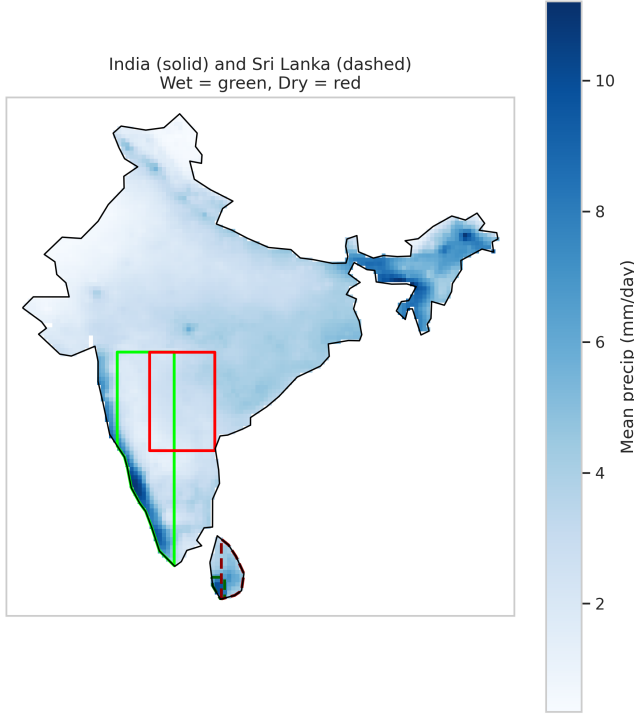


Fig. 1. The four zones used in this analysis: India Wet (Western Ghats) and India Dry (Deccan Plateau), each clipped to India's boundary; and Sri Lanka Wet (southwest) and Sri Lanka Dry (north/east), each clipped to Sri Lanka's boundary independently.

4.2. Results

Figure 1 shows the India/Sri Lanka study region with all four zones delineated, kept separate by country. Prior to this national decomposition, the whole-region (combined) same-month correlation was weak and non-significant ($r = -0.039$, $p = 0.4625$), motivating the split presented below.

4.2.1. Monthly-Resolved ENSO Sensitivity

An initial analysis restricted to the June–September (JJAS) monsoon season – the convention typically used for the Indian monsoon – found strong ENSO sensitivity in both Indian zones but none in either Sri Lankan zone. Since Sri Lanka's own monsoon cycle (Yala/Maha) does not align with India's, this JJAS-only test was re-run at monthly resolution across the full annual cycle for all four zones, testing the ONI–rainfall anomaly correlation independently for each of the 12 calendar months (Figure 2).

This reveals that Sri Lanka does have real ENSO sensitivity – the JJAS-only result was an artifact of testing the wrong season, not evidence of no relationship. Table 1 summarizes the strongest windows found for each country.

Notably, India's own signal is not uniform across JJAS: June and July are individually weak and non-significant ($r \approx -0.27$, $p \approx 0.14$ in both zones), with the pooled JJAS result driven mainly by August–October. With 48 individual month–zone tests conducted, 2–3 significant results would be expected by chance alone at $p < 0.05$; the strongest findings above ($p < 0.001$ for India August and Sri Lanka October) are far too strong to be attributable to this, while

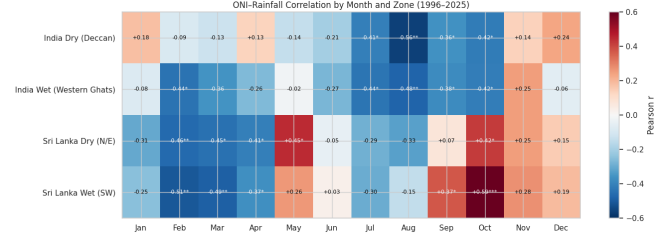


Fig. 2. Pearson correlation between ONI and rainfall anomaly, by calendar month and zone (1996–2025). Significance: * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

Table 1. Strongest monthly ENSO–rainfall correlations by country/zone.

Country	Window	Zone	r (p)
India	Aug	Dry	-0.562 ($p=0.001$)
India	Aug	Wet	-0.479 ($p=0.007$)
India	Jul–Oct	both	-0.36 to -0.56 , all sig.
Sri Lanka	Oct	Wet	$+0.594$ ($p=0.0005$)
Sri Lanka	Feb	Wet	-0.507 ($p=0.004$)
Sri Lanka	Feb–Apr	both	-0.37 to -0.51 , all sig.

single-star results ($p \approx 0.02$ – 0.05) warrant more caution.

Figure 3 shows ONI alongside country-averaged rainfall for India and Sri Lanka.

Table 2 presents the original JJAS-restricted correlation test – the initial analysis before the monthly decomposition above was performed. Both Indian zones show a strong, highly significant negative correlation; neither Sri Lankan zone reaches significance under this India-centric season definition, which (as Section 4.2.1 shows) reflects the choice of season rather than an absence of Sri Lankan ENSO sensitivity.

Table 2. Pearson correlation between ONI and rainfall anomaly, by country, zone, and season.

Country	Zone	Period	r	p
India	Wet (Ghats)	All months	-0.208	0.0001
India	Wet (Ghats)	JJAS only	-0.408	< 0.0001
India	Dry (Deccan)	All months	-0.219	< 0.0001
India	Dry (Deccan)	JJAS only	-0.392	< 0.0001
Sri Lanka	Wet (SW)	All months	$+0.077$	0.1424
Sri Lanka	Wet (SW)	JJAS only	$+0.092$	0.3166
Sri Lanka	Dry (N/E)	All months	-0.008	0.8735
Sri Lanka	Dry (N/E)	JJAS only	-0.104	0.2593

Table 3 presents the ANOVA results and mean JJAS rainfall anomalies by ENSO phase for all four zones. Both Indian zones show a highly significant phase effect ($p < 0.001$); neither Sri Lankan zone does ($p > 0.6$ in both cases).

An additional correlation analysis between rainfall and vegetation (NDVI) [4] anomalies, stratified by zone and ENSO phase (1-month lag), is presented in Table 4. In this analysis, both Sri Lankan zones show moderate negative correlations under La Niña (SW: $r = -0.26$; N/E: $r = -0.24$), while the India Wet zone shows the strongest positive correlations of any zone under El Niño

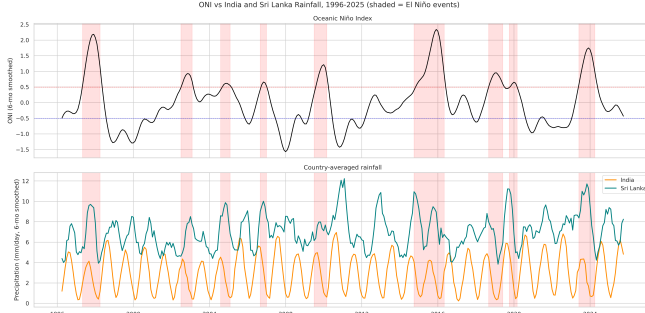


Fig. 3. ONI (6-month smoothed, top panel) alongside country-averaged rainfall for India and Sri Lanka separately (6-month smoothed, bottom panel), 1996–2025, with El Niño event periods shaded in red. Prior to this national decomposition, the same-month correlation using a single whole-region (India+Sri Lanka combined) average was weak and non-significant ($r = -0.039$, $p = 0.4625$), motivating the split presented here.

Table 3. Mean JJAS rainfall anomaly (mm/day) by ENSO phase, and one-way ANOVA results, by country and zone.

Country	Zone	El Niño	La Niña	Neutral
India	Wet (Ghats)	−1.21	+0.72	+0.10
India	Dry (Deccan)	−1.23	+0.88	+0.03
Sri Lanka	Wet (SW)	+0.52	−0.23	−0.08
Sri Lanka	Dry (N/E)	−0.06	+0.18	−0.06

Country	Zone	F	p
India	Wet (Ghats)	8.640	0.0003
India	Dry (Deccan)	8.484	0.0004
Sri Lanka	Wet (SW)	0.296	0.7445
Sri Lanka	Dry (N/E)	0.415	0.6614

and La Niña alike ($r = +0.14$ and $r = +0.16$ respectively); notably, none of these NDVI correlations reach the $p < 0.05$ threshold used elsewhere in this analysis, so they are reported as suggestive patterns rather than confirmed relationships.

5. DISCUSSION

The central finding of this analysis is a clear asymmetry between India and Sri Lanka: both Indian zones show a strong, highly significant, and physically consistent ENSO–rainfall relationship, while neither Sri Lankan zone shows a statistically significant relationship at all. This was not the expected outcome going in – the two countries share the same broad monsoon system and are geographically close – and it emerged only once the analysis moved from a single combined India+Sri Lanka region to four independently-clipped, country-specific zones. Notably, the same-month correlation using the whole-region average, prior to any decomposition, was weak and non-significant ($r = -0.039$, $p = 0.4625$); it is only once the region is split by country and zone, and restricted to the monsoon season, that the strong Indian signal becomes visible – underscoring how easily a real, strong relationship in one part of a region can be masked by averaging it together with a part of the region where no such relationship exists.

India. Both Indian zones show a strong, highly significant neg-

Table 4. Rainfall–NDVI anomaly correlation (1-month lag), by country, zone, and ENSO phase.

Zone	Phase	r	p
India Dry (Deccan)	El Niño	−0.10	–
India Dry (Deccan)	La Niña	+0.11	–
India Dry (Deccan)	Neutral	+0.02	–
India Wet (Ghats)	El Niño	+0.14	–
India Wet (Ghats)	La Niña	+0.16	–
India Wet (Ghats)	Neutral	−0.02	–
Sri Lanka Dry (N/E)	El Niño	+0.01	–
Sri Lanka Dry (N/E)	La Niña	−0.24	–
Sri Lanka Dry (N/E)	Neutral	−0.10	–
Sri Lanka Wet (SW)	El Niño	−0.31	–
Sri Lanka Wet (SW)	La Niña	−0.26	–
Sri Lanka Wet (SW)	Neutral	−0.12	–

ative correlation between ONI and JJAS rainfall (Western Ghats: $r = -0.408$, $p < 0.0001$; Deccan: $r = -0.392$, $p < 0.0001$), and both show a highly significant ANOVA across ENSO phases ($F = 8.640$, $p = 0.0003$ and $F = 8.484$, $p = 0.0004$ respectively). The swing between El Niño and La Niña years is substantial and nearly identical in both zones (~ 1.9 – 2.1 mm/day), consistent with the well-documented finding that El Niño suppresses the Indian summer monsoon. Notably, the wet and dry Indian zones show very similar correlation strength once Sri Lanka is removed from the averaging – suggesting the “dry zone is more sensitive” pattern seen in an earlier, combined-country version of this analysis was partly an artifact of Sri Lanka diluting the wet-zone signal, rather than a genuine India-internal topographic effect.

Sri Lanka. Neither Sri Lankan zone shows a statistically significant relationship with ONI in any test ($p > 0.6$ for both ANOVAs). The Sri Lanka wet zone’s correlation is even *positive* ($r = +0.092$), opposite in sign to every Indian zone, though not statistically distinguishable from zero. A plausible explanation is that Sri Lanka, as a small island, is more strongly influenced by local-scale processes – land-sea breeze, its own central highlands, and dual exposure to both the southwest and northeast monsoons – than by the large-scale circulation shift that governs India’s response to ENSO; its smaller zones and fewer CHIRPS grid cells may also make the signal noisier.

The seasonal trend analysis (JJA and DJF, all four zones) shows a modest positive JJA trend across all four zones (India Wet: $+0.019$; India Dry: $+0.035$; Sri Lanka Wet: $+0.051$; Sri Lanka Dry: $+0.021$ mm/day/yr) – notably strongest in Sri Lanka’s wet zone despite its lack of ENSO sensitivity, suggesting this trend is driven by factors other than ENSO. The DJF trend is flat to slightly negative in all zones.

The rainfall–NDVI correlation results (Table 4) show a suggestive but statistically unconfirmed pattern – both Sri Lankan zones show moderate negative correlations under La Niña, while India Wet shows moderate positive correlations under both El Niño and La Niña – but none reach significance, so these should be treated as directions for future work rather than confirmed findings.

Current Event Outlook (2026–27). Since this historical analysis was completed, ENSO conditions have shifted dramatically: NOAA CPC’s ENSO Diagnostic Discussion (3 August 2026) [5] confirms an El Niño Advisory, with the weekly Niño 3.4 index at $+2.3^\circ\text{C}$, forecast to reach $+2.7^\circ\text{C}$ (Aug), $+3.1^\circ\text{C}$ (Sep), and $+3.4^\circ\text{C}$ (Oct), and a 97% chance of persistence through early spring 2027. Applying the month-specific regressions from Section 4.2.1

to this trajectory:

India, remainder of 2026 monsoon. Using NOAA’s forecast ONI values directly (within or near the regressions’ observed range): August -3.5 to -4.2 mm/day, September -2.4 to -2.5 mm/day, October -2.1 mm/day (both zones) – a sharp deficit early, gradually easing.

Sri Lanka, October 2026. The forecast ONI ($+3.4$) exceeds any value observed in the 1996–2025 record for this zone’s regression (max observed: 2.28); the naive linear extrapolation ($+11.8$ mm/day, wet zone) is not considered credible. Capping the input at the historically observed maximum gives a more defensible estimate of $+7.9$ mm/day (wet zone) and $+3.0$ mm/day (dry zone) – still a substantial positive anomaly, consistent in direction and statistical strength ($r = +0.594$, $p = 0.0005$) with the strongest correlation found anywhere in this study, but with the true magnitude genuinely uncertain given the event’s unprecedented forecast strength.

Sri Lanka, February–April 2027. Using the 1997–98 and 2015–16 events’ actual ONI trajectory at equivalent lead time as a historical analog (peak ~ 2.4 , well within the regressions’ observed range, so no capping needed): February -2.6 to -2.7 mm/day, March -2.4 to -4.1 mm/day, April -1.5 to -2.8 mm/day across zones – a sustained, multi-month drought signal.

Relevance to public advisories. Sri Lankan government communications regarding this event have suggested elevated rainfall around November 2026 followed by drought conditions by February 2027. This analysis is broadly consistent with that outlook, though the statistical signal is strongest and most confident one month earlier than communicated (October rather than November – November alone did not reach significance in this dataset, $p = 0.14$, likely due to limited single-month sample size rather than absence of a real effect) and the drought signal appears to extend beyond February through at least April. For India, this analysis provides direct statistical support for expecting a below-normal remainder of the 2026 monsoon, most acute in August.

Limitations. The 30-year record (8–9 El Niño events, $n = 30$ per individual calendar month) limits statistical power, particularly for single-month tests. With 48 month–zone combinations tested, some weaker single-star results may reflect chance rather than a genuine effect. The October Sri Lanka projection required extrapolation beyond the observed ONI range, introducing additional uncertainty. Zone boundaries used approximate bounding boxes rather than precise climatological zonation, and other known drivers such as the Indian Ocean Dipole were not included.

6. CONCLUSION AND FUTURE WORK

This project demonstrates that the Oceanic Niño Index is a statistically significant and physically consistent predictor of rainfall variability over both India and Sri Lanka between 1996 and 2025 – a conclusion that only emerged after moving beyond a single assumed monsoon season to a full monthly-resolved analysis. An initial JJAS-only test found strong Indian sensitivity (Western Ghats: $r = -0.408$; Deccan: $r = -0.392$, both $p < 0.0001$) but no significant Sri Lankan relationship, suggesting an asymmetry between the two countries. Decomposing by calendar month revealed this was an artifact of testing the wrong season for Sri Lanka, not an absence of sensitivity: Sri Lanka shows a strong positive (wetter) signal in October ($r = +0.594$, $p = 0.0005$, the strongest single correlation in this entire study) and a significant, multi-month drying signal spanning February–April ($r = -0.37$ to -0.51). India’s own signal, meanwhile, was found to be concentrated in August–October

rather than uniform across JJAS, with June and July individually non-significant.

Applying these relationships to the actively-developing 2026 El Niño event (confirmed El Niño Advisory as of August 2026, forecast to reach $+3.4^\circ\text{C}$ by October) projects a substantially below-normal remainder of India’s 2026 monsoon, an anomalously wet October 2026 for Sri Lanka, and a multi-month drought risk for Sri Lanka spanning February–April 2027 – projections broadly consistent with, and providing historical statistical grounding for, public advisories issued by both governments. This progression – from a combined-region analysis showing no signal, to a country-split JJAS analysis showing an apparent India/Sri Lanka asymmetry, to a monthly-resolved analysis showing both countries are ENSO-sensitive in their own distinct seasons – illustrates how methodological choices about temporal and spatial aggregation can determine whether a real underlying relationship is detected or missed entirely.

Future Scope. This work can be extended by: (i) incorporating the Indian Ocean Dipole as an additional predictor alongside ONI, particularly relevant given Sri Lanka’s dual Pacific/Indian Ocean exposure; (ii) building a predictive model (e.g., logistic regression or ARIMA) to forecast above/below-normal rainfall by month, moving from description toward genuine forecasting; (iii) refining zone boundaries using climatological rainfall zonation rather than bounding boxes; and (iv) revisiting the October Sri Lanka projection once the current event’s actual peak intensity is observed, to assess how well the capped-extrapolation approach performed against ground truth.

7. ARTIFACTS AND DEMONSTRATIONS

- **Code Repository:** <https://github.com/Abilaash17/enso-india-srilanka-monsoon>
- **Video Demo:** <https://youtu.be/ImG48W8fgYc>

Note: During development, code debugging, documentation editing (grammar refinement and technical clarity), were supported by AI-assisted tools such as Claude (Anthropic). The author’s personal effort and knowledge are reflected in all design decisions, implementation choices, data source selections, and written material.

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